

29.5 Selection bias and honest inference

Everything in Part III assumed that the model was fixed before the data were seen. After a search, the reported model, its coefficients, its error estimate and its test statistics are the ones that looked best, and each is biased: effects look larger, errors smaller, and intervals cover less often than they claim. This section proves the direction of the bias, measures it, and describes three remedies.

29.5.1 The winner's curse

Proposition 29.5.1. What selection does to estimates and intervals

- (a) (*Selected maxima are biased upwards.*) Let $Z_1, \dots, Z_m$ have finite means $\theta_1, \dots, \theta_m$, and let $\hat{J}$ be an index at which $Z_{\hat{J}} = \max_j Z_j$. Then

$$E(Z_{\hat{J}}) \geq \max_j \theta_j \geq E(\theta_{\hat{J}}),$$

so $E(Z_{\hat{J}} - \theta_{\hat{J}}) \geq 0$. If the Z_j are independent $N(\theta, 1)$ with a common mean and $m \geq 2$, the bias is $E \max_j Z_j - \theta > 0$.

- (b) (*Selected minima of risk estimates are optimistic.*) Let $\hat{R}_k$ be an unbiased estimate of R_k for each candidate k in a finite set, and let $\hat{k}$ minimize $\hat{R}_k$. Then

$$E(\hat{R}_{\hat{k}}) \leq \min_k R_k \leq E(R_{\hat{k}}).$$

- (c) (*Selected intervals under-cover.*) Let $Z \sim N(\theta, 1)$, and report the interval $Z \pm z_{\alpha/2}$ only when $|Z| > c$. If $c \geq z_{\alpha/2}$, the probability that the reported interval covers θ , given that it is reported, is 0 at $\theta = 0$. It tends to $1 - \alpha$ as $|\theta| \rightarrow \infty$.

Proof.

- (a) $\max_j Z_j \geq Z_i$ for every i , so $E \max_j Z_j \geq \theta_i$ for every i , which is the first inequality. The second holds pointwise: $\theta_{\hat{J}} \leq \max_j \theta_j$. In the normal case, $\max_j Z_j - \theta = \max_j (Z_j - \theta)$ has the distribution of the largest of $m \geq 2$ independent standard normals, which exceeds $Z_1 - \theta$ with positive probability and is never smaller, so its mean exceeds $E(Z_1 - \theta) = 0$.
- (b) $\hat{R}_{\hat{k}} \leq \hat{R}_k$ for every k ; take expectations to get $E \hat{R}_{\hat{k}} \leq R_k$ for every k . The second inequality holds pointwise.
- (c) At $\theta = 0$ the interval covers iff $|Z| \leq z_{\alpha/2}$, which is incompatible with $|Z| > c \geq z_{\alpha/2}$. As $|\theta| \rightarrow \infty$, $\Pr(|Z| > c) \rightarrow 1$, so the conditional coverage tends to the unconditional $\Pr(|Z - \theta| \leq z_{\alpha/2}) = 1 - \alpha$. ■

Part (a) is the **winner's curse**: the largest of ten unbiased estimates of equal means overstates its mean by 1.539 standard errors on average, and the coefficient and t statistic of a regressor that “won” a screening are biased in the same way. Part (b) applies to every criterion of this chapter, since C_p and cross-validation are unbiased for each fixed candidate (Proposition 29.2.1(a) and Proposition 29.3.1); the search degrees of freedom of [Section 29.1](#) measure the same effect.

Example. Freedman's paradox

Freedman (1983) described the following procedure, which many analyses follow in some form. Fit $n = 100$ cases on 50 regressors, none of which is related to the response. Keep those with p -value below 0.25, refit on the survivors, and report. In one simulated data set 15 regressors survive the first pass, 7 of them are significant at 5% in the refit, and the overall F test of the refit has p -value 0.0030. Over 2000 data sets, an average of 12.3 regressors survive and 5.6 are significant, and the overall F test rejects at 5% in a fraction 0.94 of them. Pure noise produces a model that every standard test endorses.

```
import numpy as np
from scipy import stats
rng = np.random.default_rng(2913)
n, m = 100, 50

def ols(X, y):
    """Coefficients, t statistics (slopes), overall F p-value; X without intercept.
    """
    A = np.column_stack([np.ones(len(y)), X])
    b, *_ = np.linalg.lstsq(A, y, rcond=None)
    e = y - A @ b
    df = len(y) - A.shape[1]
    s2 = e @ e / df
    t = b / np.sqrt(s2 * np.diag(np.linalg.inv(A.T @ A)))
    ssr = np.sum((A @ b - y.mean()) ** 2)
    F = (ssr / X.shape[1]) / s2
    return t[1:], df, stats.f.sf(F, X.shape[1], df)

def screen_and_refit():
    X = rng.normal(size=(n, m))
    y = rng.normal(size=n)
    t, df, _ = ols(X, y)
    keep = np.abs(t) > stats.t.ppf(1 - 0.25 / 2, df)
    t2, df2, p_F = ols(X[:, keep], y)
    survivors
    return keep.sum(), np.sum(np.abs(t2) > stats.t.ppf(0.975, df2)), p_F

kept, signif, p_F = screen_and_refit()
print(f"one data set: {kept} kept, {signif} significant at 5%, overall F p-value {
    p_F:.4f}")
runs = np.array([screen_and_refit() for _ in range(500)])
print("over 500 data sets: mean kept", runs[:, 0].mean(), " mean significant", runs
   [:, 1].mean(),
    " Pr(F significant)", np.mean(runs[:, 2] < 0.05))
```

[Run this code online](#)

29.5.2 Coverage after selection

The failure in [Proposition 29.5.1\(c\)](#) concerns intervals reported only after selection. A subtler failure affects intervals for a coefficient that is always reported, when the *model* around it was chosen from the data.

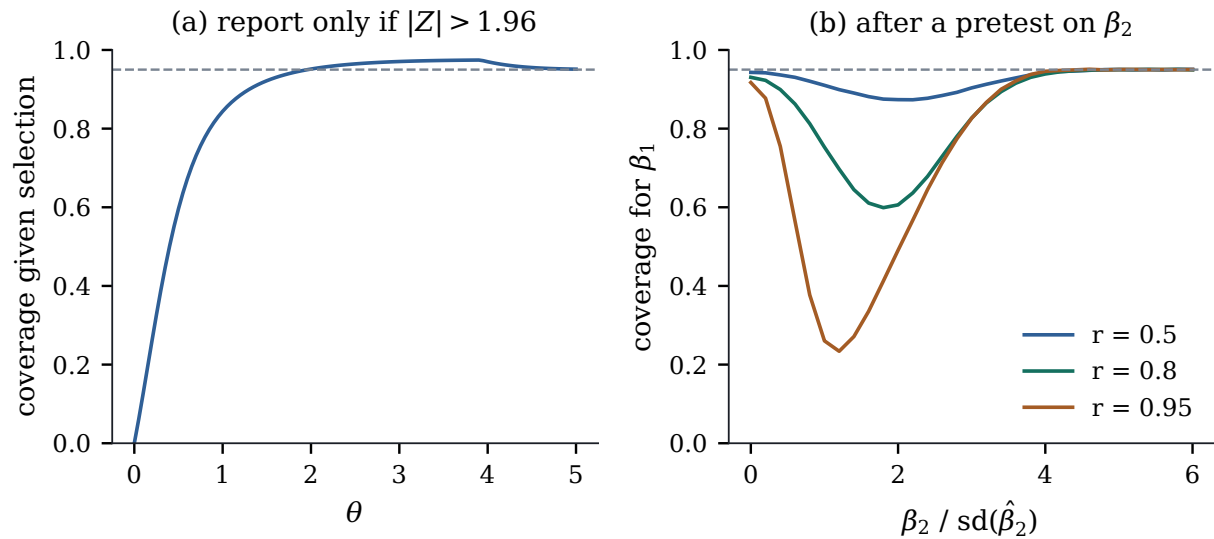


Figure 29.5.1: (a) Coverage of $Z \pm 1.96$, $Z \sim N(\theta, 1)$, given that it is reported only when $|Z| > 1.96$. (b) Coverage for β_1 of the usual 95% interval from the model chosen by a 5% test of $\beta_2 = 0$, for three correlations between the regressors (200000 simulations per point).

Example. Coverage after a pretest

Take two standardized regressors with correlation r , known σ , and the question: what is β_1 ? The analyst first tests $\beta_2 = 0$ at level 5%. If the test rejects, she reports the usual 95% interval for β_1 from the model with both regressors; otherwise, the interval from the model with $\mathbf{x}_1$ alone. Each interval is correct for its own model. Panel (b) of Figure 29.5.1 shows the coverage of the reported interval as a function of $\delta = \beta_2 / \text{sd}(\hat{\beta}_2)$, the standardized size of the coefficient being tested. When $\delta = 0$ the short model is correct and coverage is close to nominal. When δ is large the test always keeps $\mathbf{x}_2$ and coverage is nominal. In between, the test often drops $\mathbf{x}_2$ although the short slope is biased by $r\beta_2$ (Theorem 19.1.2), and the coverage falls to a minimum of 0.873 (at $\delta = 2.2$) for $r = 0.5$, 0.599 (at $\delta = 1.8$) for $r = 0.8$ and 0.234 (at $\delta = 1.2$) for $r = 0.95$.

Panel (a) is the one-dimensional case of Proposition 29.5.1(c) with $c = 1.96$: the conditional coverage is 0 at $\theta = 0$, 0.844 at $\theta = 1$, 0.951 at $\theta = 2$ and 0.971 at $\theta = 3$, where it slightly over-covers.

```
z = stats.norm.ppf(0.975)

def conditional_coverage(theta, c=z):
    """Pr(|Z - theta| <= z | |Z| > c) for Z ~ N(theta, 1): the reported interval."""
    lo, hi = theta - z, theta + z
    inside_sel = (stats.norm.cdf(max(hi, c) - theta) - stats.norm.cdf(max(lo, c) -
    theta)) \
        + (stats.norm.cdf(min(hi, -c) - theta) - stats.norm.cdf(min(lo, -c) - theta)
    )
    return inside_sel / (stats.norm.sf(c - theta) + stats.norm.cdf(-c - theta))

def pretest_coverage(delta, r, reps=200_000):
    """Coverage of the usual 95% interval for beta_1 after a 5% pretest of beta_2 =
    0.
```

```

Two standardized regressors with correlation r, sigma known; delta = beta_2 / sd
(beta_2 hat)."""
sd2 = 1 / np.sqrt(1 - r**2)                                # sd of either long-
model slope                                                # covariance of (b1, b2)
C = np.array([[1, -r], [-r, 1]]) / (1 - r**2)              # in the long model
b = rng.normal(size=(reps, 2)) @ np.linalg.cholesky(C).T + [0.0, delta * sd2]
keep = np.abs(b[:, 1] / sd2) > z                            # the pretest keeps x2
b1_short = b[:, 0] + r * b[:, 1]                            # short slope (sd 1)
cover_long = np.abs(b[:, 0]) <= z * sd2                     # true beta_1 = 0
cover_short = np.abs(b1_short) <= z
return np.mean(np.where(keep, cover_long, cover_short))

for r, d_min in ((0.5, 2.2), (0.8, 1.8), (0.95, 1.2)):      # the minimizing
    delta of the figure
    print(f"r = {r}: coverage at delta = 0, {d_min}, 5:",
          ", ".join(f"{pretest_coverage(d, r, 50_000):.3f}" for d in (0.0, d_min,
5.0)))

```

[Run this code online](#)

The coverage depends on the unknown δ , and it is worst where the data are least able to tell whether $\mathbf{x}_2$ belongs. No simple correction repairs this: Leeb and Pötscher (2005, 2006) showed that the distribution of a post-selection estimator cannot be estimated consistently, uniformly in the parameters, so bootstrapping the whole procedure does not rescue the naive interval either.

29.5.3 Honest inference, I: splitting the sample

The simplest remedy separates the two uses of the data. Choose the model on one part of the sample, and compute intervals from the other part, as if the model had been fixed in advance, which, for that part, it was (Cox 1975).

Proposition 29.5.2. Sample splitting

Let $\mathbf{Y} = (\mathbf{Y}_1^\top, \mathbf{Y}_2^\top)^\top$, where $\mathbf{Y}_2 \sim N_{n_2}(\boldsymbol{\mu}_2, \sigma^2 \mathbf{I})$ is independent of $\mathbf{Y}_1$, and let $\mathbf{X}_2$ be the fixed design of the second part. Let $\hat{M}$ be any rule that chooses a set of columns from $\mathbf{Y}_1$ (and the design), with values in the set of M for which $\mathbf{X}_{2,M}$ has full column rank $p_M < n_2$. For each such M , let $\boldsymbol{\beta}_M = (\mathbf{X}_{2,M}^\top \mathbf{X}_{2,M})^{-1} \mathbf{X}_{2,M}^\top \boldsymbol{\mu}_2$, and let $I_{M,j}$ be the usual level $1 - \alpha$ t interval for the j th coefficient from the least squares fit of $\mathbf{Y}_2$ on $\mathbf{X}_{2,M}$. Then, given $\mathbf{Y}_1$, $I_{\hat{M},j}$ covers $\beta_{\hat{M},j}$ with probability at least $1 - \alpha$, with equality if $\boldsymbol{\mu}_2 \in \mathcal{C}(\mathbf{X}_{2,\hat{M}})$. The same holds unconditionally.

Proof. Given $\mathbf{Y}_1$, the model $M = \hat{M}$ is fixed and $\mathbf{Y}_2$ still has its distribution, by independence. For fixed M , write $\mathbf{M}_M$ for the projection onto $\mathcal{C}(\mathbf{X}_{2,M})$ and $w = [(\mathbf{X}_{2,M}^\top \mathbf{X}_{2,M})^{-1}]_{jj}$. The estimate $\hat{\beta}_{M,j}$ is normal with mean $\beta_{M,j}$ and variance $\sigma^2 w$, so $Z = (\hat{\beta}_{M,j} - \beta_{M,j}) / (\sigma \sqrt{w}) \sim N(0, 1)$. The residual sum of squares $\|(\mathbf{I} - \mathbf{M}_M) \mathbf{Y}_2\|^2$ is a function of $(\mathbf{I} - \mathbf{M}_M) \mathbf{Y}_2$, which is uncorrelated with $\mathbf{M}_M \mathbf{Y}_2$ and hence independent of it (Theorem 3.4.1), and $W = \|(\mathbf{I} - \mathbf{M}_M) \mathbf{Y}_2\|^2 / \sigma^2 \sim \chi^2(\nu, \gamma)$ with $\nu = n_2 - p_M$ and $\gamma = \|(\mathbf{I} - \mathbf{M}_M) \boldsymbol{\mu}_2\|^2 / \sigma^2$ (Theorem 4.3.2(a)). The interval covers iff $|Z| \leq t_{\alpha/2}(\nu) \sqrt{W/\nu}$, iff $W > \nu Z^2 / t_{\alpha/2}(\nu)^2$. Conditioning on Z and using Proposition 4.1.3, this probability is nondecreasing in γ , and at $\gamma = 0$ it is $1 - \alpha$ by the definition of the t distribution. Averaging over $\mathbf{Y}_1$ gives the unconditional statement.

The target $\beta_{M,j}$ is the coefficient of $\mathbf{x}_j$ in the best approximation to the mean by the chosen columns, which is what the fitted coefficient estimates, right model or wrong. The costs: selection sees only part of the data, the intervals are wider, and the result depends on the random split. Wasserman and Roeder (2009) and Rinaldo, Wasserman and G'Sell (2019) develop the idea for high-dimensional and assumption-lean settings.

29.5.4 Honest inference, II: covering every model

The second remedy keeps all the data and widens the intervals until they cover the coefficients of every model that could have been chosen, simultaneously. Then it does not matter which one was.

Proposition 29.5.3. Simultaneous inference over submodels

Assume the normal model Eq. 7.3.1 with $\text{rank}(\mathbf{X}) = P < n$, let s^2 be the full-model estimate with $\nu = n - P$, and let $\mathcal{M}$ be a collection of column subsets M with $\mathbf{X}_M$ of full column rank. For $M \in \mathcal{M}$ and $j \in M$ let $\mathbf{v}_{M,j} = \mathbf{X}_M(\mathbf{X}_M^\top \mathbf{X}_M)^{-1} \mathbf{e}_j$, where $\mathbf{e}_j$ picks out the coefficient of $\mathbf{x}_j$, and

$$\beta_{M,j} = \mathbf{v}_{M,j}^\top \mathbf{X} \boldsymbol{\beta}, \quad \hat{\beta}_{M,j} = \mathbf{v}_{M,j}^\top \mathbf{Y},$$

and for $K > 0$ let

$$I_{M,j}(K) = [\hat{\beta}_{M,j} \pm K s \|\mathbf{v}_{M,j}\|].$$

Let N be the number of pairs (M, j) and q the dimension of the span of the $\mathbf{v}_{M,j}$.

- (a) With $K = \sqrt{q F_\alpha(q, \nu)}$, $\Pr(\beta_{M,j} \in I_{M,j}(K) \text{ for all } M \in \mathcal{M}, j \in M) \geq 1 - \alpha$.
- (b) The same holds with $K = t_{\alpha/(2N)}(\nu)$.
- (c) If K has the property in (a), then for every selection rule $\hat{M}$ with values in $\mathcal{M}$, a function of $\mathbf{Y}$ that may also depend on the unknown $\boldsymbol{\beta}$ and σ , $\Pr(\beta_{\hat{M},j} \in I_{\hat{M},j}(K) \text{ for all } j \in \hat{M}) \geq 1 - \alpha$.
- (d) The smallest K with the property in (c) for every such rule, including rules that depend on the unknown parameters (the convention of Berk, Brown, Buja, Zhang and Zhao 2013), is K_{PoSI} , the $1 - \alpha$ quantile of $\max_{M,j} |\mathbf{v}_{M,j}^\top (\mathbf{Y} - \mathbf{X} \boldsymbol{\beta})| / (s \|\mathbf{v}_{M,j}\|)$, whose distribution does not depend on $\boldsymbol{\beta}$ or σ .

Proof. In each case $\hat{\beta}_{M,j} - \beta_{M,j} = \mathbf{v}_{M,j}^\top (\mathbf{Y} - \mathbf{X} \boldsymbol{\beta})$ with $\mathbf{v}_{M,j} \in \mathcal{C}(\mathbf{X})$. (a) Let $\mathcal{V}$ be the span of the $\mathbf{v}_{M,j}$, a subspace of $\mathcal{C}(\mathbf{X})$ of dimension q . By Theorem 13.2.2(a), with probability exactly $1 - \alpha$, $|\mathbf{v}^\top (\mathbf{Y} - \mathbf{X} \boldsymbol{\beta})| \leq \sqrt{q F_\alpha(q, \nu)} s \|\mathbf{v}\|$ for every $\mathbf{v} \in \mathcal{V}$, in particular for every $\mathbf{v}_{M,j}$. (b) Each $(\hat{\beta}_{M,j} - \beta_{M,j}) / (s \|\mathbf{v}_{M,j}\|)$ has the $t(\nu)$ distribution, because $\mathbf{v}_{M,j}^\top \mathbf{Y}$ is independent of s^2 when $\mathbf{v}_{M,j} \in \mathcal{C}(\mathbf{X})$ (Theorem 7.5.1). Apply Theorem 13.4.1(b) with $\alpha_{M,j} = \alpha/N$. (c) The event in (c) contains the simultaneous event of (a). (d) The simultaneous event of (a) is $\{\max_{M,j} T_{M,j} \leq K\}$ with $T_{M,j} = |\mathbf{v}_{M,j}^\top (\mathbf{Y} - \mathbf{X} \boldsymbol{\beta})| / (s \|\mathbf{v}_{M,j}\|)$. Its distribution is free of $\boldsymbol{\beta}$ and σ because $\mathbf{Y} - \mathbf{X} \boldsymbol{\beta} = \sigma \mathbf{z}$ with $\mathbf{z} \sim N_n(\mathbf{0}, \mathbf{I})$ and $s = \sigma \|(\mathbf{I} - \mathbf{M})\mathbf{z}\| / \sqrt{\nu}$. So K_{PoSI} has the property in (a) and hence in (c). If $K < K_{\text{PoSI}}$, take the rule that selects a model containing a pair at which $T_{M,j}$ is largest, a rule that uses $\boldsymbol{\beta}$ and so is allowed in (c); its coverage is $\Pr(\max T_{M,j} \leq K) < 1 - \alpha$.

This is the PoSI (post-selection inference) approach of Berk, Brown, Buja, Zhang and Zhao (2013): model selection treated as a multiple-comparison problem of [Chapter 13](#). The Scheffé bound never exceeds $\sqrt{PF_\alpha(P, \nu)}$, however many models are searched, and the price is paid whether or not the search was aggressive.

Example. Honest intervals for the state regressions

For the state data, with the intercept always in, the 32 candidate models contain $N = 80$ slope coefficients. Every $\mathbf{v}_{M,j}$ is orthogonal to $\mathbf{1}$, so they span at most $q = 5$ dimensions, and $\nu = 44$. The multipliers are 2.015 for the naive t interval, 3.12 for PoSI (by simulation), 3.484 for Scheffé and 3.684 for Bonferroni. In the model chosen by AIC, single parenthood and urbanization, the t statistics computed with the full-model s are 5.46 and 1.83. Single parenthood clears every multiplier. Urbanization clears none, not even the naive one: AIC kept it because its squared t exceeds about 2 (the threshold of [Section 29.2](#)), which is a decision about prediction, not evidence of an effect.

```
import itertools
import statsmodels.api as sm
rng = np.random.default_rng(2914)
data = sm.datasets.statelcrime.load_pandas().data.drop(index="District of Columbia")
names = ["hs_grad", "poverty", "single", "white", "urban"]
ys = np.log(data["violent"].to_numpy())
Xs = np.column_stack([np.ones(len(ys)), data[names].to_numpy()])
ns, P = Xs.shape
nu = ns - P

# unit vectors u with coefficient of x_j in submodel M = (u' y) / ||v||, for
# every M and j in M
U = []
for k in range(1, 6):
    for M in itertools.combinations(range(1, 6), k):
        XM = Xs[:, [0, *M]]
        V = XM @ np.linalg.inv(XM.T @ XM)
        # column i is v for
        # coefficient i
        for i in range(1, len(M) + 1):
            U.append(V[:, i] / np.linalg.norm(V[:, i]))
U = np.array(U)
# 80 unit vectors, all
# in C(X) minus 1
Qb, _ = np.linalg.qr(Xs)
Qb = Qb[:, 1:]
# orthonormal basis of
# C(X) orthogonal to 1
W = U @ Qb
# coordinates of the u'
# s in that basis
sims = 100_000
Zp = rng.normal(size=(sims, P - 1))
s = np.sqrt(rng.chisquare(nu, size=sims) / nu)
K_posi = np.quantile(np.max(np.abs(Zp @ W.T), axis=1) / s, 0.95)
K_t = stats.t.ppf(0.975, nu)
K_bonf = stats.t.ppf(1 - 0.025 / len(U), nu)
K_sch = np.sqrt((P - 1) * stats.f.ppf(0.95, P - 1, nu))
print(f"{len(U)} coefficients; multipliers: t {K_t:.3f}, PoSI {K_posi:.3f}, "
      f"Bonferroni {K_bonf:.3f}, Scheffe {K_sch:.3f}")
```

[Run this code online](#)

Example. Three ways to report after forward selection

Take $n = 100$ cases of ten normal regressors with pairwise correlations 0.5, $\sigma = 1$, choose a model by forward selection with AIC, and report 95% intervals for the chosen coefficients, with the projection coefficients $\beta_{\hat{M},j}$ as targets. Over 3000 data sets, the fraction of reported intervals that cover their targets is:

	naive	split in half	Scheffé
no signal: coverage	0.440	0.946	1.000
no signal: mean half-width	0.228	0.351	0.516
three slopes of 0.3: coverage	0.868	0.956	1.000
three slopes of 0.3: mean half-width	0.248	0.379	0.559

With no signal, fewer than half of the naive intervals cover: a coefficient is reported only because its estimate was large. Splitting restores the nominal level at the cost of intervals about one and a half times as wide. The Scheffé intervals, with multiplier $\sqrt{10F_{0.05}(10, 89)} = 4.403$, are valid for every model and every rule, and here they are very conservative, because forward selection explores only a small part of the 5120 coefficients they protect.

29.5.5 Honest inference, III: conditioning on the selection

The third remedy uses all the data and asks for coverage *given* that the model was selected. In the one-dimensional setting of [Proposition 29.5.1\(c\)](#) this can be done exactly.

Example. A selective interval

Let $Z \sim N(\theta, 1)$ be reported only when $|Z| > 1.96$, and suppose $Z = 2.5$ is observed. Given selection, Z has the normal distribution truncated to $|z| > 1.96$. Let F_θ be its distribution function. Then $F_\theta(Z)$ is uniform on $(0, 1)$ given selection, whatever θ , and $F_\theta(z)$ decreases in θ ([Exercise \(C1\)](#)). So $\{\theta : 0.025 \leq F_\theta(2.5) \leq 0.975\}$ is an interval with conditional coverage exactly 95% for every θ . Numerically it is $[-0.42, 4.33]$, against the naive $[0.54, 4.46]$. An estimate just past the threshold is weak evidence once the threshold is taken into account, and the selective interval includes zero.

```
from scipy.optimize import brentq

def trunc_cdf(x, theta, c=z):
    """Distribution function at x of N(theta, 1) conditioned on |Z| > c."""
    den = stats.norm.sf(c - theta) + stats.norm.cdf(-c - theta)
    if x <= -c:
        num = stats.norm.cdf(x - theta)
    elif x < c:
        num = stats.norm.cdf(-c - theta)
    else:
        num = stats.norm.cdf(-c - theta) + stats.norm.cdf(x - theta) - stats.norm.
        cdf(c - theta)
```



```

return num / den

z_obs = 2.5
lower = brentq(lambda th: trunc_cdf(z_obs, th) - 0.975, -10, z_obs)
upper = brentq(lambda th: trunc_cdf(z_obs, th) - 0.025, -10, 10)
print(f"naive interval [{z_obs - z:.2f}, {z_obs + z:.2f}], selective interval [{
    lower:.2f}, {upper:.2f}]")

```

[Run this code online](#)

Lee, Sun, Sun and Taylor (2016) extended the construction to selection events that are polyhedra $\{\mathbf{A}\mathbf{Y} \leq \mathbf{b}\}$, which covers the lasso at a fixed penalty, forward stepwise selection (Tibshirani, Taylor, Lockhart and Tibshirani 2016) and thresholding: a linear function of $\mathbf{Y}$, given the selection and the part of $\mathbf{Y}$ orthogonal to it, is a truncated normal. The intervals can be very long near the selection boundary. Selecting on part of the data but using all of it for inference (“data carving”, Fithian, Sun and Taylor 2014) shortens them. In high dimensions the debiased lasso of Theorem 28.5.1 gives intervals for the full model’s coefficients instead.

None of the three routes is free. The cheapest honest inference is still to fix the model before seeing the data, and to use selection for prediction, where its errors can be measured.

29.5.6 Exercises

29.5.6.1 A. Check your understanding

Exercise. A1

In [Proposition 29.5.1\(c\)](#), take $c = 1.5 < z_{0.025}$. Compute the conditional coverage at $\theta = 0$.

29.5.6.2 B. Practice

Exercise. B1

In the orthonormal design of [Proposition 29.2.5](#) with $\sigma = 1$ and a zero coefficient, show that $E(|Z_j| \mid |Z_j| > c) = \phi(c)/(1 - \Phi(c))$. Evaluate it for $c = \sqrt{2}$ and $c = 1.96$, and interpret it as the average size of a selected null coefficient.

Solution. By symmetry $E(|Z| \mid |Z| > c) = E(Z \mid Z > c) = \int_c^\infty z\phi(z) dz / (1 - \Phi(c)) = \phi(c)/(1 - \Phi(c))$. For $c = \sqrt{2}$ this is $0.1468/0.0786 = 1.87$; for $c = 1.96$ it is $0.0584/0.0250 = 2.34$. A coefficient that is truly zero, once selected, is reported with an estimate of about two standard errors on average: the winner’s curse of [Proposition 29.5.1\(a\)](#), in its purest form.

Exercise. B2

Verify that the 32 candidate models of [Example \(Honest intervals for the state regressions\)](#) contain 80 slope coefficients, and that every $\mathbf{v}_{M,j}$ for a slope is orthogonal to $\mathbf{1}$. Why does this let Scheffé’s bound use $q = 5$ instead of 6?

Solution. Each of the 5 regressors appears in $2^4 = 16$ of the models, so there are $5 \times 16 = 80$ pairs. The vector $\mathbf{v}_{M,j} = \mathbf{X}_M(\mathbf{X}_M^\top \mathbf{X}_M)^{-1} \mathbf{e}_j$ satisfies $\mathbf{X}_M^\top \mathbf{v}_{M,j} = \mathbf{e}_j$, and the intercept column of $\mathbf{X}_M$ is $\mathbf{1}$, with $\mathbf{e}_j$ zero in that position, so $\mathbf{1}^\top \mathbf{v}_{M,j} = 0$. All the $\mathbf{v}_{M,j}$ lie in $\mathcal{C}(\mathbf{X}) \cap \mathbf{1}^\perp$, of dimension 5, and Theorem 13.2.2(a) needs only a subspace containing them.

29.5.6.3 C. Going deeper

Exercise. C1

Let F_θ be the distribution function of $N(\theta, 1)$ truncated to a set A of positive probability. Show that for $\theta_1 < \theta_2$ the density ratio $f_{\theta_2}(z)/f_{\theta_1}(z)$ is increasing in $z \in A$, and deduce that $F_{\theta_2}(z) \leq F_{\theta_1}(z)$ for every z . Why does this make the set in [Example \(A selective interval\)](#) an interval?

Solution. On A , $f_\theta(z) = \phi(z - \theta)/\Pr_\theta(A) = \phi(z)e^{\theta z - \theta^2/2}/\Pr_\theta(A)$, so the ratio is a positive constant times $e^{(\theta_2 - \theta_1)z}$, increasing in z . Put $g = f_{\theta_2}/f_{\theta_1}$ on A . If $g(z) \leq 1$ then $g \leq 1$ on $A \cap (-\infty, z]$, so $F_{\theta_2}(z) \leq F_{\theta_1}(z)$. If $g(z) > 1$ then $g > 1$ on $A \cap [z, \infty)$, so $1 - F_{\theta_2}(z) \geq 1 - F_{\theta_1}(z)$. Either way $F_{\theta_2}(z) \leq F_{\theta_1}(z)$. For fixed z , $\theta \mapsto F_\theta(z)$ is therefore nonincreasing (and continuous), so $\{\theta : 0.025 \leq F_\theta(z) \leq 0.975\}$ is an interval.

Exercise. C2

In [Example \(Coverage after a pretest\)](#), let $\hat{\beta}_1^S$ be the short slope and $\hat{\beta}_2$ the long-model estimate of β_2 . Use Lemma 19.1.1 to show that the two are independent, and write the coverage of the reported interval as

$$\Pr(|T_2| \leq z) \Pr(|\hat{\beta}_1^S - \beta_1| \leq z \text{sd}(\hat{\beta}_1^S)) + \Pr(|T_2| > z, |\hat{\beta}_1 - \beta_1| \leq z \text{sd}(\hat{\beta}_1)),$$

where $T_2 = \hat{\beta}_2/\text{sd}(\hat{\beta}_2)$. Explain why only the first term depends on the bias $r\beta_2$ of the short slope in this simple way.